

Comparative Analysis of Normalization and Cross-Validation Scenarios for Deep Learning-Based Dengue Fever Classification

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Abstract—

Index Terms—Deep Learning, Dengue Fever, Artificial Neural Network, Classification, Normalization, Cross Validation

I. INTRODUCTION

Dengue Fever is one of the most significant mosquito-borne infectious diseases in tropical and subtropical countries. According to the *World Health Organization (WHO)*, the global incidence of Dengue has increased rapidly over the past decades, particularly in Southeast Asia and other tropical regions. Early detection and classification of Dengue Fever are essential to reduce mortality rates and improve clinical treatment effectiveness [1].

Conventional diagnosis of Dengue Fever generally relies on clinical symptoms and laboratory examinations performed by healthcare professionals. Common symptoms include fever, headache, nausea, muscle pain, and skin rash, while laboratory indicators such as platelet count, hematocrit, leukocyte count, and hemoglobin level are often used to support diagnosis. However, manual interpretation of clinical data may require considerable time and may be affected by human error, especially when the number of patients increases significantly [2].

Recent developments in Artificial Intelligence (AI) and Deep Learning have demonstrated promising performance in medical classification and prediction systems. Deep Learning models are capable of learning complex non-linear relationships from medical datasets and have been widely applied in disease diagnosis, image classification, and healthcare decision support systems [3]. Among various Deep Learning approaches, Artificial Neural Network (ANN) models have shown good capability in binary classification problems due to their ability to capture hidden patterns within clinical datasets [4].

Several previous studies have implemented Machine Learning and Deep Learning techniques for Dengue prediction and classification. However, many studies mainly focus on model performance without comprehensively analyzing the influence of preprocessing techniques such as data normalization and

validation strategies. In fact, normalization techniques can significantly affect Deep Learning performance because numerical feature scales may influence the convergence process during model training [5]. Furthermore, Cross Validation is important to evaluate model stability and generalization capability in healthcare datasets [6].

Therefore, this study proposes a Deep Learning-based Dengue Fever classification system using clinical symptom and laboratory examination datasets. This research focuses on comparative analysis of normalization techniques and Cross Validation scenarios to determine the optimal classification performance. Two dataset scenarios were evaluated, including a full dataset consisting of symptom and laboratory features, and a symptom-only dataset. Three normalization methods were implemented, namely without normalization, StandardScaler, and MinMaxScaler, resulting in six experimental scenarios. In addition, K-Fold Cross Validation was applied to evaluate the robustness and stability of the proposed model.

The results of this study are expected to contribute to the development of healthcare decision support systems and provide insights regarding the influence of preprocessing techniques on Deep Learning-based medical classification systems.

A. Research Contributions

The main contributions of this study are summarized as follows:

- Comparative analysis of normalization techniques for Dengue Fever classification.
- Comparative analysis between full dataset and symptom-only dataset.
- Evaluation of model stability using K-Fold Cross Validation.
- Development of a Deep Learning-based healthcare classification system.
- Investigation of the influence of clinical symptoms and laboratory examination features on classification performance.

- Evaluation of the effectiveness of StandardScaler and MinMaxScaler normalization techniques in improving Deep Learning performance.
- Analysis of model generalization capability through multiple cross-validation scenarios.
- Identification of the best experimental scenario based on classification performance metrics including Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
- Development of an experimental framework that can be applied to other healthcare classification problems using Deep Learning approaches.

II. RELATED WORK

TABLE I
SUMMARY OF RELATED STUDIES

Study	Method	Accuracy
Study A	ANN	92%
Study B	CNN	94%
Study C	Random Forest	89%

III. RESEARCH METHODOLOGY

A. Research Workflow

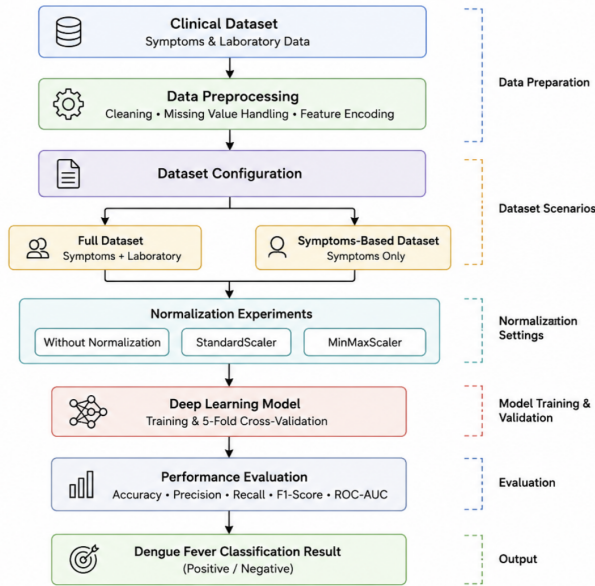


Fig. 1. Research Workflow

B. Dataset

The dataset used in this study consists of clinical symptom and laboratory examination data.

TABLE II
DATASET ATTRIBUTE CATEGORIES

Category	Number of Features
Symptoms	9
Laboratory Features	4
Total Features	13

C. Dataset Scenarios

Two dataset scenarios were evaluated in this study.

TABLE III
DATASET SCENARIOS

Scenario	Dataset Composition
Scenario 1	Symptoms + Laboratory Features
Scenario 2	Symptoms Features Only

D. Data Preprocessing

The preprocessing stage includes:

- Data Cleaning
- Missing Value Handling
- Feature Encoding
- Data Normalization

E. Normalization Experiments

Three normalization methods were evaluated:

- Without Normalization
- StandardScaler
- MinMaxScaler

TABLE IV
NORMALIZATION EXPERIMENT SCENARIOS

Experiment	Dataset	Normalization
Exp-1	Full Dataset	None
Exp-2	Full Dataset	StandardScaler
Exp-3	Full Dataset	MinMaxScaler
Exp-4	Symptoms Only	None
Exp-5	Symptoms Only	StandardScaler
Exp-6	Symptoms Only	MinMaxScaler

F. Proposed Deep Learning Model

TABLE V
ANN ARCHITECTURE

Layer	Neurons	Activation
Input Layer	13	-
Dense Layer 1	64	ReLU
Dropout Layer	-	-
Dense Layer 2	32	ReLU
Output Layer	1	Sigmoid

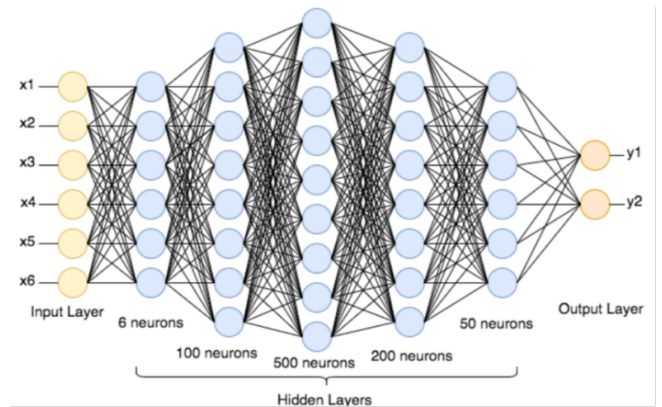


Fig. 2. Artificial Neural Network Architecture

G. Cross Validation

K-Fold Cross Validation with K=5 was implemented to evaluate model stability and generalization capability.

TABLE VI
CROSS VALIDATION SCENARIOS

Scenario	Dataset	Normalization
CV-1	Full Dataset	None
CV-2	Full Dataset	StandardScaler
CV-3	Full Dataset	MinMaxScaler
CV-4	Symptoms Only	None
CV-5	Symptoms Only	StandardScaler
CV-6	Symptoms Only	MinMaxScaler

H. Evaluation Metrics

The evaluation metrics used in this study include:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

IV. RESULTS AND DISCUSSION

A. Training Performance

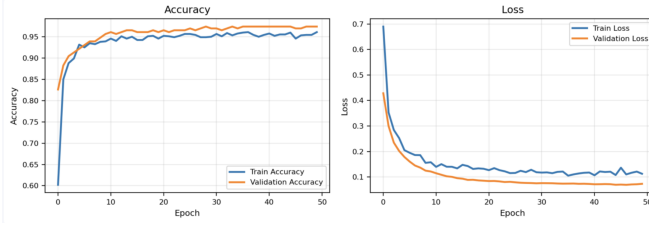


Fig. 3. Training Accuracy and Loss

B. Normalization Experiment Results

TABLE VII
EXPERIMENTAL RESULTS

Experiment	Accuracy	Precision	Recall	F1-Score
Exp-1				
Exp-2				
Exp-3				
Exp-4				
Exp-5				
Exp-6				

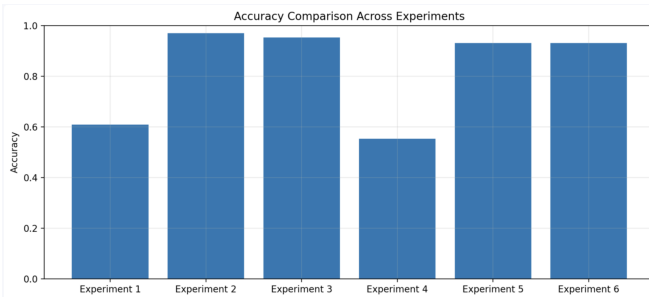


Fig. 4. Comparison of Experimental Accuracy

C. Cross Validation Results

TABLE VIII
CROSS VALIDATION RESULTS

Scenario	Mean Accuracy	Std Deviation
CV-1		
CV-2		
CV-3		
CV-4		
CV-5		
CV-6		

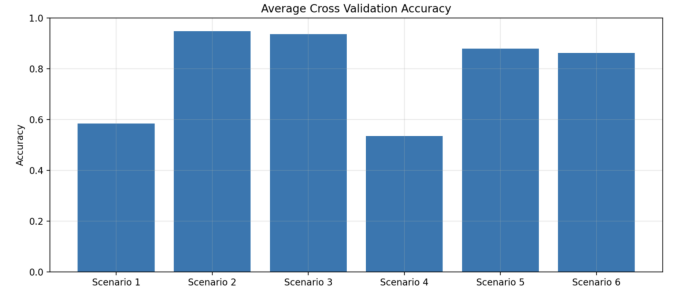


Fig. 5. Cross Validation Accuracy Comparison

D. Confusion Matrix Analysis

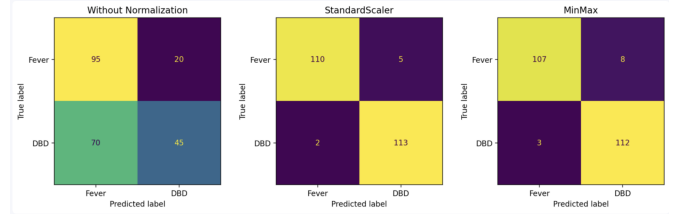


Fig. 6. Confusion Matrix Comparison for Three Normalization Methods

E. ROC Curve Analysis

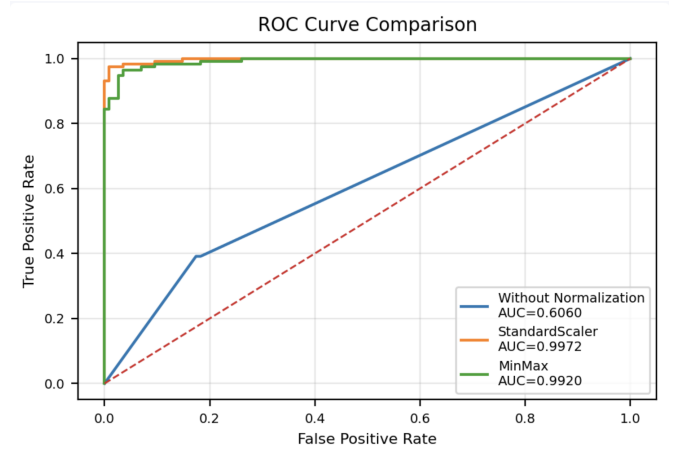


Fig. 7. ROC Curve Comparison for Three Normalization Methods

F. Discussion

V. CONCLUSION

VI. FUTURE WORK

ACKNOWLEDGMENT

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